

Yeshwantrao Chavan College of engineering, Nagpur.

Data Analytics Lab 03

Name :	Mrunali Jibhakate	Class/roll no :	6 th EE B - 107
Reg. No :	23071350		
Aim :	To demonstrate and implement data grouping, aggregation functions, and horizontal concatenation of multiple datasets using the Python pandas library.,		
Implementation Steps :	Part A: Grouping and Aggregation • Data Definition: • Grouping • Aggregation. Part B: Dataframe Concatenation • Multiple Data Sources • Horizontal Stacking		
Program Code :	<pre>1) import numpy as np # linear algebra import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv) # Input data files are available in the read-only "../input/" directory # For example, running this (by clicking run or pressing Shift+Enter) will list all files under the input directory import os for dirname, _, filenames in os.walk('/kaggle/input'): for filename in filenames: print(os.path.join(dirname, filename)) 2) data1 = {'Name': ['Jai', 'Anuj', 'Jai', 'Princi', 'Gaurav', 'Anuj', 'Princi', 'Abhi'], 'Age': [27, 24, 22, 32, 33, 36, 27, 32] 'Address': ['Nagpur', 'Kanpur', 'Allahabad', 'Kannuaj', 'Jaunpur', 'Kanpur', 'Allahabad', 'Aligarh'], 'Qualification': ['Msc', 'MA', 'MCA', 'Phd', 'B.Tech', 'B.com', 'Msc', 'MA']}</pre> 3) df = pd.DataFrame(data1) grp1 = df.groupby('Name') grp1.agg(np.sum) 4) # Define a dictionary containing employee data data1 = {'key': ['K0', 'K1', 'K2', 'K3'], 'Name': ['Jai', 'Princi', 'Gaurav', 'Anuj'], 'Age': [27, 24, 22, 32]} # Define a dictionary containing employee data data2 = {'key': ['K0', 'K1', 'K2', 'K3'],		

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	<pre>'Address':['Nagpur', 'Kanpur', 'Allahabad', 'Kannuaj'], 'Qualification':['Btech', 'B.A', 'Bcom', 'B.hons']} # Convert the dictionary into DataFrame df = pd.DataFrame(data1) # Convert the dictionary into DataFrame df1 = pd.DataFrame(data2) display(df, df1) 5) res = pd.concat([df, df1], axis=1) res</pre>																																																																											
Output :	2]: <table><thead><tr><th></th><th></th><th>Age</th><th>Address</th><th>Qualification</th></tr></thead><tbody><tr><td colspan="5">Name</td></tr><tr><td></td><td>Abhi</td><td>32</td><td>Aligarh</td><td>MA</td></tr><tr><td></td><td>Anuj</td><td>60</td><td>KanpurKanpur</td><td>MAB.com</td></tr><tr><td></td><td>Gaurav</td><td>33</td><td>Jaunpur</td><td>B.Tech</td></tr><tr><td></td><td>Jai</td><td>49</td><td>NagpurAllahabad</td><td>MscMCA</td></tr><tr><td></td><td>Princi</td><td>59</td><td>KannuajAllahabad</td><td>PhdMsc</td></tr></tbody></table> <table><thead><tr><th></th><th>key</th><th>Name</th><th>Age</th></tr></thead><tbody><tr><td>0</td><td>K0</td><td>Jai</td><td>27</td></tr><tr><td>1</td><td>K1</td><td>Princi</td><td>24</td></tr><tr><td>2</td><td>K2</td><td>Gaurav</td><td>22</td></tr><tr><td>3</td><td>K3</td><td>Anuj</td><td>32</td></tr></tbody></table> <table><thead><tr><th></th><th>key</th><th>Address</th><th>Qualification</th></tr></thead><tbody><tr><td>0</td><td>K0</td><td>Nagpur</td><td>Btech</td></tr><tr><td>1</td><td>K1</td><td>Kanpur</td><td>B.A</td></tr><tr><td>2</td><td>K2</td><td>Allahabad</td><td>Bcom</td></tr><tr><td>3</td><td>K3</td><td>Kannuaj</td><td>B.hons</td></tr></tbody></table>			Age	Address	Qualification	Name						Abhi	32	Aligarh	MA		Anuj	60	KanpurKanpur	MAB.com		Gaurav	33	Jaunpur	B.Tech		Jai	49	NagpurAllahabad	MscMCA		Princi	59	KannuajAllahabad	PhdMsc		key	Name	Age	0	K0	Jai	27	1	K1	Princi	24	2	K2	Gaurav	22	3	K3	Anuj	32		key	Address	Qualification	0	K0	Nagpur	Btech	1	K1	Kanpur	B.A	2	K2	Allahabad	Bcom	3	K3	Kannuaj	B.hons
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Result :	This lab successfully demonstrates how to consolidate data. Grouping is essential for summarizing large datasets based on categories, while Concatenation allows for the integration of data from different sources into a unified structure.																																			

Marks :		Name of Lecturer : Dr. S.A. Dhondse
		Signature :